Lvzhou Chen

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Employment

2022–current **Assistant Professor**, *Purdue University*, USA.

2020–2022 **R. H. Bing Instructor**, *The University of Texas at Austin*, USA.

Education

2014–2020 **Ph.D. in Mathematics**, *The University of Chicago*, USA.
Advisor: Danny Calegari
Thesis: Surfaces in graphs of groups and the stable commutator length

2015 **M.S. in Mathematics**, *The University of Chicago*, USA.

2014 **B.S. in Mathematics and Applied Mathematics**, *Fudan University*, China.
Thesis Advisors: Zhi Lü and Yijun Yao
Thesis: $\mathbb{Z}_2$-cohomological rigidity of small covers over n-Löbell

Research Interests

Geometry, topology, geometric/combinatorial group theory, and dynamics in low dimensions

Publications

15. **(with Nicolaus Heuer) Uniform spectral gap of scl in 2-orbifolds**, *submitted*, arXiv:2604.21059, 20 pages.
14. **(with Yash Lodha) The Wiegold problem and free products of left-orderable groups**, *submitted*, arXiv:2510.26073, 36 pages.
13. **(with Nicolaus Heuer) Spectral gap of scl in graphs of groups and 3-manifolds**, *Amer. J. Math.*, to appear, arXiv: 1910.14146, 50 pages.
12. **The Kervaire conjecture and the minimal complexity of surfaces**, *Trans. Amer. Math. Soc.*, **379** (2026), 587–626.
11. **(with Sebastian Hurtado, Homin Lee) A height gap in $\mathrm{GL}_d(\overline{\mathbb{Q}})$ and almost laws**, *Groups Geom. Dyn.*, **19** (2025), no. 3, 899–912.
10. **(with Alexander J. Rasmussen) Irrational rotations and 2-filling rays**, *Ergodic Theory and Dynamical Systems*, **45** (2025), no. 6, 1673–1697.
9. **(with Nicolaus Heuer) Stable commutator length in right-angled Artin and Coxeter groups**, *J. Lond. Math. Soc.*, **107** (2023), no. 1, 1–60.
8. **(with Alexander J. Rasmussen) Laminations and 2-filling rays on infinite type surfaces**, *Ann. Inst. Fourier*, **73** (2023), no. 6, 2305–2369.

7. **(with Chloe I. Avery) Stable torsion length**, *Int. Math. Res. Not. (IMRN)*, (2023), no. 16, 13817–13866.
6. **(with Danny Calegari) Normal subgroups of big mapping class groups**, *Trans. Amer. Math. Soc. Ser. B*, **9** (2022), 957–976.
5. **(with Danny Calegari) Big mapping class groups and rigidity of the simple circle**, *Ergodic Theory and Dynamical Systems*, **41** (2021), no. 7, 1961–1987.
4. **(with Santana Afton, Danny Calegari, Rylee Alanza Lyman) Nielsen realization for infinite-type surfaces**, *Proc. Amer. Math. Soc.*, **149** (2021), no. 4, 1791–1799.
3. **Scl in graphs of groups**, *Invent. Math.*, **221** (2020), no. 2, 329–396.
2. **Spectral gap of scl in free products**, *Proc. Amer. Math. Soc.*, **146** (2018), no.7, 3143–3151.
1. **Scl in free products**, *Algebr. Geom. Topol.*, **18** (2018), no.6, 3279–3313.

Awards and Honors

2026 **Kamil Duszenko Award**, *Wrocław Mathematicians Foundation*, Poland.
granted for outstanding work or research that has significantly contributed to the deepening of knowledge and further progress in the field of mathematics

2026 **Ruth & Joel Spira Teaching Award**, *Purdue University*, USA.
For excellence in undergraduate and graduate teaching

2025-2028 **Principal Investigator**, *NSF Grant DMS 2506702*, USA.
Topological minimal surfaces and applications to topology and group theory

2020 **Wirszup Fellowship**, *University of Chicago*, USA.
Given to an excellent finishing graduate student

Invited Talks

June 2026 **Advanced School at ICMAT, Madrid, Spain**.

May 2026 **International Young Seminar on Bounded Cohomology and Simplicial Volume**.

April 2026 **Topology Seminar, UC Berkeley**.

April 2026 **Topology Seminar, UT Austin**.

March 2026 **Topology Seminar, Notre Dame**.

March 2026 **Groups and Dynamics Seminar, Princeton/IAS**.

Feb. 2026 **Topology Seminar, Rice**.

Feb. 2026 **Dynamics Seminar, IU Indianapolis**.

Feb. 2026 **Automorphic Forms and Representation Theory Seminar, Purdue**.

Jan. 2026 **Geometric Group Theory Seminar, Vanderbilt**.

Dec. 2025 **Topology and Geometric Group Theory Seminar, OSU**.

June 2025 **2K-GATE Workshop I, KAIST and KIAS, Korea**.

Nov. 2024 **Geometry and Topology Seminar, MIT**.

July 2024 **Shanghai Geometric Group Theory Workshop 2024, Shanghai Center for Mathematical Sciences**.

May 2024 **Geometric Group Theory Seminar, Cambridge University**.

April 2024 **Low Dimensional Actions Conference, Inst. Henri Poincaré, Paris**.

March 2024 **Dynamics Seminar, IUPUI**.

Sept. 2023 **Geometry Seminar, Lehigh**.

May 2023 **Geometry and Analysis on Groups research seminar, Vienna**.

May 2023 **Online Seminar on Bounded Cohomology and Simplicial Volume**.

May 2023 **Knot Online Seminar (K-OS)**.

April 2023 **Colloquium, University of Hawaii at Manoa**.

March 2023 **Loo-Keng Hua Forum for Young Mathematicians, Academy of Mathematics and Systems Science, Chinese Academy of Sciences**.

Nov. 2022 **Topology Seminar, Notre Dame**.

Oct. 2022 **Geometry and Topology Seminar, Caltech**.

Aug. 2022 **Wasatch Topology Conference 2022**.

Aug. 2022 **Geometric Group Theory Summer School, Shanghai Center for Mathematical Sciences**.

Aug. 2022 **Metric Geometry for Young Scholars, Capital Normal University**.

March 2022 **Semi-plenary talk, 2022 Spring Topology and Dynamics Conference**.

March 2022 **Colloquium, Purdue University**.

Nov. 2021 **Geometric Analysis Seminar, Purdue University**.

Nov. 2021 **CMSA Interdisciplinary Science Seminar, Harvard University**.

June 2021 **International Young Seminar on Bounded Cohomology and Simplicial Volume**.

June 2021 **GGT Seminar, University of Münster**.

June 2021 **Infinite-type surfaces and big mapping class groups session, NCNGT Conference**.

June 2021 **Oberseminar Groups and Geometry, Universität Bielefeld Bielefeld**.

May 2021 **Topology Festival, Cornell University**.

May 2021 **GGT session, 2021 Spring Topology and Dynamics Conference**.

March 2021 **Geometry Topology Seminar, Georgia Tech**.

Nov. 2020 **Big Surfaces Seminar, online**.

Oct. 2020 **Topology/Geometry Seminar, Rutgers**.

Sept. 2020 **GGT Seminar, Ohio State University**.

Aug. 2020 **Topology Seminar, University of Texas at Austin**.

June 2020 **Hyperbolic Lunch, University of Toronto**.

June 2020 **Hyperbolic geometry and manifolds session, NCNGT Conference**.

Feb. 2020 **ANT-CoG Seminar, University of North Carolina at Greensboro**.

Feb. 2020 **Geometry Seminar, University of Michigan**.

Jan. 2020 **Geometry and Topology Seminar, Caltech**.

Dec. 2019 **Topology Seminar, Fudan University**.

Oct. 2019 **Geom/Top Seminar, Washington University in St. Louis**.

Oct. 2019 **Geometry and Topology Seminar, University of Chicago**.

Sept. 2019 **Dynamics Seminar, Boston College**.

March 2019 **Topology and Geometric Group Theory Seminar, Cornell University**.

Nov. 2018 **Fall AMS southeastern sectional meeting, University of Arkansas**.

Sept. 2017 **Fall AMS eastern sectional meeting, University at Buffalo**.

Referee Experience

J. AMS, Invent. Math. (3 times), GAFA, Adv. Math. (3 times), M. EMS, J. Topol., T. AMS, Math. Z. (3 times), Comment. Math. Helv., J. LMS, Algebr. Geom. Topol. (4 times), P. AMS, BLMS, J. Topol. Anal. (twice), Israel J Math, L'Enseign Math, New York J. Math, Topology and its Applications, Ann. Fenn. Math.

Teaching Experience (as Instructors)

Purdue

Spring 2026 **MA 351**, *Elementary Linear Algebra (two sections)*.

Fall 2025 **MA 341**, *Foundations Of Analysis*.

Spring 2025 **MA 351**, *Elementary Linear Algebra*.

Spring 2025 **MA 697**, *Mapping class groups (grad. topics course, joint with Sam Nariman)*.

Fall 2024 **MA 661**, *Modern Differential Geometry*.

Fall 2024 **MA 697**, *Bounded Cohomology (grad. topics course, joint with Sam Nariman)*.

Fall 2023 **MA 351**, *Elementary Linear Algebra*.

Spring 2023 **MA 351**, *Elementary Linear Algebra (two sections)*.

UT Austin

Spring 2022 **M 392 C**, *The Gromov norm and bounded cohomology (grad. topics course)*.

Fall 2021 **M 408 S**, *Integral Calculus for Science*.

Spring 2021 **M 427 L**, *Advanced Calculus for Applications II*.

Fall 2020 **M 328 K**, *Introduction to Number Theory*.

UChicago

2019–2020 **Math 152 and 153**, *Calculus*.

2018–2019 **Math 152 and 153**, *Calculus*.

2017–2018 **Math 152 and 153**, *Calculus*.

2016–2017 **Math 151, 152 and 153**, *Calculus*.

Service and Mentorship

Spring 2026 **Mentor of the Purdue Experimental Math Lab**, *Purdue*.
Mentoring undergrads for a project exploring topological methods in group theory

2026–current **Co-organizer of the Purdue Geometric Group Theory Seminar**.
Invited and hosted speakers

2025–current **Advisor of Purdue Undergraduate Math Club**.
Suggested some activities

2022–current **Co-organizer of the Purdue Geometry and Geometric Analysis Seminar**.
Invited and hosted speakers

2022–current **Advising PhD students**, *Purdue*.
Geoffrey Baring (current)

2022–current **Mentoring undergraduate students**, *Purdue*.
Minseung Son (Point-set Topology, Spring 2024 DRP; Faculty Mentor since Spring 2025)

2023–current **Advising the Organization of Purdue Directed Reading Program**, *Purdue*.
Provide guidance to graduate students organizing the DRP, including suggestions on the format and timeline of the program, pairing mentors and mentees, finding new organizers

Spring 2025 **Co-organizer of Geometry Groups and Dynamics Workshop**.
Suggested speakers, advertised it, sought funding, and made arrangements to host it at Purdue

Spring 2024 **Co-organizer of Geometry Groups and Dynamics Workshop**.
Suggested speakers, advertised it, and partially supported it with funding

Spring 2023 **Co-organizer of the Midwest Topology Seminar, Purdue**.
Suggested several speakers, contributed ideas about the structure of the conference, advertised it, and partially supported it with funding

2020–2022 **Co-organizer of the Topology Seminar at UT Austin**.
Suggested, invited and hosted speakers

Fall 2021 **Mentor of the Texas Geometry Lab**.
Guided a group of three undergraduates (Simon Xiang, Jimmy Xin, Ruiqi Zou) to explore stable commutator length in free groups using computer experiments from a topological/combinatorial point of view

Spring 2020 **Organizer of Geometric Group Theory session in NCNGT conference**.
Designed mini-sessions, invited speakers and hosted the session online

Fall 2018 **Organizer of Reading Group**, *on surface subgroups*.
Divided papers into manageable parts for one-hour talks, assigned talks to participating postdocs and graduate students, gave several talks

2017–2018 **Mentor for Directed Reading Program**.
Found suitable topics and textbooks for undergraduate students to study
- Mentee: Mary Stelow. Topic: 1-dimensional Complex Dynamics
- Mentee: Jeremy Atos. Topic: Fundamental Groups and Homology Groups